

2024. E. OHIKOA

A3 $f(x) = x^3 + Ax^2 + Bx + C$

- zureu ukitzarleak $x = -1$ eta $x = 2$ -rako paraleloak

Paraleloak badira moldo berdino daukie, beraz $f'(x_0)$ berdino daukie. $\rightarrow \boxed{f'(-1) = f'(2)}$

- $x = 1$ denean MUTUR ERLANBOA dauko.

Mutur erlatibetara $f'(x_0) = 0$ beraz:

$$\boxed{f'(1) = 0}$$

$$f(0) = \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} = \left(\frac{0}{0} \right) \stackrel{\text{L'Hopital}}{=} \lim_{x \rightarrow 0} \frac{2 \cdot e^{2x}}{1} = 2.$$

Beraz $\boxed{f(0) = 2}$

Kalkulatuko da $f'(x)$ eta emandako baldintzak ordetkatu:

$$f(x) = x^3 + Ax^2 + Bx + C$$

$$f'(x) = 3x^2 + 2Ax + B$$

$$f'(-1) = f'(2) \rightarrow 3 - 2A + B = 12 + 4A + B$$

$$-9 = 6A \rightarrow A = -\frac{9}{6} \rightarrow \boxed{A = -\frac{3}{2}}$$

$$f'(1) = 0 \rightarrow 3 + 2A + B = 0 \rightarrow 3 + 2 \cdot \left(-\frac{3}{2}\right) + B = 0$$

$$\rightarrow \boxed{B = 0}$$

$$f(0) = 2 \rightarrow 0 + A \cdot 0 + B \cdot 0 + C = 2 \rightarrow \boxed{C = 2}$$

- b) fren grafikoreu zureu ukitzarilearen ek.
 $x = -1$ quen puntu etorko, $A = -3$, $B = 0$ eta
 $C = 4$ -rako.

$f(x) = x^3 - 3x^2 + 4$. Izaukle, zureu ukitzarilearren
ekuazioa $y = y_0 + m(x - x_0)$ da.

$$\left. \begin{array}{l} y_0 = f(-1) \\ m = f'(-1) \end{array} \right\} \text{ lortuko da.}$$

$$y_0 = f(-1) = (-1)^3 - 3 \cdot (-1)^2 + 4 = 0. \Rightarrow \boxed{y_0 = 0}$$

$$f'(x) = 3x^2 - 6x$$

$$f'(-1) = 3(-1)^2 - 6(-1) = 9 \longrightarrow \boxed{m = 9}$$

Beraz:

$$y = 0 + 9 \cdot (x - (-1))$$

$$\boxed{y = 9x + 9}$$

B3 $f(x) = 2x e^{-2x^2}$ Domain = $\mathbb{R}$.

2024
E2 OKIKO

a) GORAKOR etc BEHERAKORTASUN TARTAK.

$$f'(x) = 2 \cdot e^{-2x^2} + 2x \cdot (-4x) \cdot e^{-2x^2}$$

$$= e^{-2x^2} (2 - 8x^2)$$

$$f'(x) = 0 \rightarrow e^{-2x^2} (2 - 8x^2) = 0 \begin{cases} e^{-2x^2} \neq 0 \\ 2 - 8x^2 = 0 \rightarrow x = \pm \frac{1}{2} \end{cases}$$

$f'(x)$	$f'(x) < 0$	$f'(x) > 0$	$f'(x) < 0$
$f(x)$	$\searrow$	$\nearrow$	$\searrow$
	$-1/2$	$1/2$	

G.T. $(-1/2, 1/2)$

B.T. $(-\infty, -1/2) \cup (1/2, +\infty)$

b) KUTUR ERLATIBOAK

Maximotau $f''(x_0) < 0$

Minimotau $f''(x_0) > 0$.

$$f''(x) = -4x e^{-2x^2} (2 - 8x^2) + e^{-2x^2} (-16x)$$

$$f''(x) = e^{-2x^2} (-8x + 32x^3 - 16x)$$

$$f''(x) = e^{-2x^2} (32x^3 - 24x)$$

$$\left\{ \begin{aligned} f''(1/2) &= e^{-2 \cdot \frac{1}{4}} (32(\frac{1}{2})^3 - 24 \cdot \frac{1}{2}) = \frac{1}{\sqrt{e}} (4 - 12) = \frac{-8}{\sqrt{e}} \text{ Max.} \\ f(1/2) &= 2 \cdot \frac{1}{2} e^{-2(1/2)^2} = e^{-1/2} \end{aligned} \right.$$

$$\left\{ \begin{aligned} f''(-1/2) &= e^{-2 \cdot 1/4} (-32(\frac{1}{2})^3 + \frac{24}{2}) = \frac{1}{\sqrt{e}} (-4 + 12) = \frac{8}{\sqrt{e}} \text{ Min.} \\ f(-1/2) &= e^{-2 \cdot 1/4} \cdot 2(-1/2) = -1/\sqrt{e} \end{aligned} \right.$$

Maximotau. $(\frac{1}{2}, e^{-1/2})$ puntuan $f''(1/2) < 0$ dabko

Minimotau $(-\frac{1}{2}, -\frac{1}{\sqrt{e}})$ puntuan $f''(-1/2) > 0$ dabko

c) aurkitu f-re asintotak

$\text{Dom } f = \mathbb{R}$ berot ez dauko asintota bertikalik.

$\lim_{x \rightarrow \infty} f(x)$ kalkulatu

$$\Rightarrow \lim_{x \rightarrow +\infty} 2x \cdot e^{-2x} = \lim_{x \rightarrow +\infty} \frac{2x}{e^{2x}} = \frac{+\infty}{+\infty} = 0$$

Infinituak konparatu $y = e^{2x}$ ren infinituak
mala joko infinituak da $y = 2x$ -ren
baino.

Beraz $x \rightarrow +\infty$ denetan ASINTOTA HORIZONTALA

$$y = 0.$$

$$\Rightarrow \lim_{x \rightarrow -\infty} 2x \cdot e^{-2x} = \lim_{x \rightarrow +\infty} 2(-x) e^{-2(-x)^2} =$$

$$= \lim_{x \rightarrow +\infty} \frac{-2x}{e^{2x^2}} = \frac{-\infty}{+\infty} = 0 \rightarrow \text{ASINT. HORIZ. } y = 0.$$

Ondorioz: Ez dauko asint. bertikale
baino bai asint. horizontale $y = 0$.